

E-ITS Asset Modeling Lifecycle and Validation Rules

This guide describes the database and service layer (Backend) logical controls that determine **when** it is permissible to assign E-ITS modules to a target object/asset (*sihtobjekt / vara*) to perform modeling.

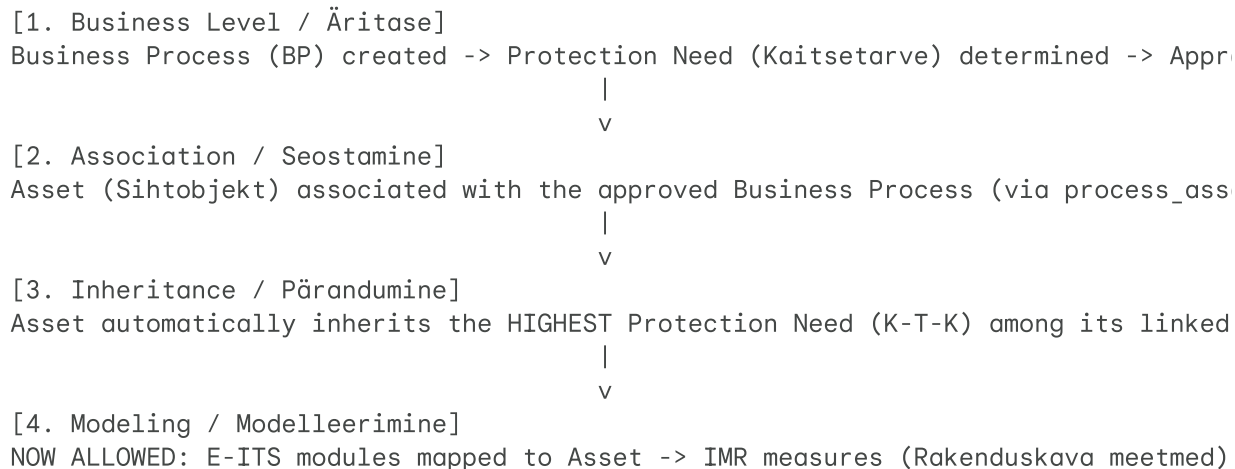
1. E-ITS Terminology Mapping Directory (Sõnastik)

To ensure the coding agent uses precise terminology in UI labels, API error responses, and audit logs, refer to this official mapping:

English Term	Estonian Term (E-ITS)	Database/Code Variable
Information Security Management System (ISMS)	Infoturbe halduse süsteem	isms
Estonian Information Security Standard (E-ITS)	Eesti infoturbestandard	eits
Business Process (BP)	Äriprotsess	business_process / bp
Target Object / Asset	Sihtobjekt / IT-vara	asset
Structural Analysis	Struktuuranalüüs	structural_analysis
Protection Need (C-I-A)	Kaitsetarve (K-T-K: Konfidentsiaalsus, Terviklus, Käideldavus)	protection_need (confidentiality , integrity , availability)
Protection Need Inheritance	Kaitsetarbe pärandumine	protection_need_inheritance
Scope Modeling	Kaitseala modelleerimine	scope_modeling
Mode of Protection	Turbeviis (põhi-, standard-, tuumikuturve)	mode_of_protection (BASELINE , STANDARD , CORE)
Infosec Measures Implementation Plan (IMR)	Infoturbe meetmete rakenduskava (IMR)	imr / imr_items
Security Measure	Turvameede	measure / eits_catalog_measure
Evidence / Proof	Asitõend / Tõendusmaterjal	evidence
Top Management / CISO	Tippjuhtkond / Infoturbejuht	approved_by (role: management / ciso)

2. Logical Step-by-Step Lifecycle (Elutsükkel)

E-ITS is a **business-process-centric standard (äriprotsessikeskne standard)**. The security measures of technical assets always derive from the specific business processes (*äriprotsessid*) those assets support. Therefore, it is mandatory to enforce the following lifecycle:



3. When CAN an Asset Be Modeled? (Validation Checklist)

Mapping an E-ITS module to an asset (modeling / *modelleerimine*) is **ONLY** allowed when the following conditions are met:

1. **Association with a Business Process exists (Struktuuranalüüsi seos olemas):** An asset must not exist in isolation. It must be associated with at least one active business process via the association table (e.g., `process_assets`).
2. **Business Process Protection Need is Approved (Äriprotsessi kaitsetarve on kinnitatud):** All business processes linked to the asset must have completed their protection need assessment, and it must be marked as **Approved** (*kinnitatud* with an `approved_by` signature in the `protection_need_summaries` table).
3. **Asset Protection Need is Calculated (Vara kaitsetarve on tuletatud):** Because the E-ITS catalog measures are filtered based on the tenant's mode of protection (*turbeviis*) and the asset's inherited protection need (*pärandunud kaitsetarve*), the asset's inherited level must be fully resolved before modeling begins.

4. Backend Validation SOP (Service Layer Logic)

Your OpenCode agent must add the following validation block at the beginning of the `ModelingService.map_module_to_target` function.

Note: Use precise Estonian-language error messages for audit logs and UI feedback to align with local compliance audits.

Service Validation Code Example (Python / FastAPI):

```

def validate_asset_ready_for_modeling(db: Session, tenant_id: uuid.UUID, asset_id:
    """
    E-ITS Rule: Validates whether an asset (sihtobjekt) is ready for modeling (mo
    The asset must be linked to at least one business process with an approved pr
    """

```

```

# 1. Check for active business process relations (Struktuuranalüüs)
process_relations = db.query(ProcessAsset).filter(ProcessAsset.asset_id == as
if not process_relations:
    raise HTTPException(
        status_code=status.HTTP_400_BAD_REQUEST,
        detail="Tõrge: Seda vara ei saa modelleerida, sest see on seostamata
    )

# 2. Verify that the protection needs of all linked business processes are ap
for relation in process_relations:
    # Fetch the protection need summary assessment for each associated proces
    summary = db.query(ProtectionNeedSummary).filter(
        ProtectionNeedSummary.business_process_id == relation.business_proces
        ProtectionNeedSummary.tenant_id == tenant_id
    ).first()

    # If the assessment is missing, or not yet approved (approved_by field is
    if not summary or not summary.approved_by:
        raise HTTPException(
            status_code=status.HTTP_400_BAD_REQUEST,
            detail=f"Tõrge: Seotud äriprotsessi '{relation.business_process.n
        )

# If all lifecycle checks pass, the asset is ready for modeling (Modelleerimi
return True

```

5. Why This Rule Is Critical (Auditing & Compliance Perspective)

- **Prevents Redundant Work (*Hoiab ära tühja töö*):** If you allowed modeling on an asset whose associated business process protection need could still be modified or is unapproved, changing that protection need later (e.g., from *Normal/Normaalne* to *High/Suur*) would silently leave the asset without mandatory standard or high-level controls (*standard- ja kõrgmeetmed*).
- **Audit-Proof Chain of Custody (*Auditikindel seoste ahel*):** An E-ITS auditor (*audiitor*) will always verify the dependency chain: Business Process (Äriprotsess) -> Protection Need (Kaitsetarve) -> Assets (Sihtobjektid) -> Modules (Moodulid) -> Measures (IMR meetmed) . This server-side validation ensures that the generated system data is methodologically pure and compliant with RIA auditing guidelines.